Mason–Stothers sees repeated roots through a derivative

$$A = X^3 \qquad B = 1 - X^3 \qquad C = 1 \qquad A + B = C$$

$$A'B - AB' = 3X^2$$

Nonzero in characteristic 0

rad(ABC): degree 4

One root at 0, three at $X^3 = 1$

$$\max(\deg A, \deg B, \deg C) = 3 \leq 4 - 1$$

The theorem needs the derivative and nonconstant hypotheses.